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Topic: CO1

Security Features in Real-World Applications

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Security Features in Real-World Applications

1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

The security feature/service required, such as confidentiality, integrity, authentication, authorization, availability

The security mechanism used to provide the identified security feature.

The security protocol or technology used to implement the mechanism.

Briefly explain how the mechanism/protocol provides the identified security feature.

Present your findings in the following format:

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
Gmail	Authentication	Password + MFA	OAuth	Verifies the identity of the user
Online banking	Confidentiality	Encryption	HTTPS/TLS	Protects data transmitted between client and server
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2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in

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Deadline : 13.08.2026

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Success is not just about reaching the top but enjoying the climb

1. I picked five apps i use almost everyday for this –Gmail,Google Pay/UPI,WhatsApp,Amazon,and Netflix.

Application	Security Feature	Security Mechanism	Protocol/ Technology	How it provides security
Gmail	Authentication	Password+Otp/ App based MFA	OAuth2.0	Whenever i try to log in,Gmail always checks my password and then for multiple security asks me for a second factor.This makes sure that it is really me and not someone who just guessed my password.
Google Pay/UPI	Confidentiality & Integrity	Encryption+ Digital Signature	TLS +UPI PIN encryption	All the Transaction data (amount account number) is encrypted why moving from my phone to the bank server,so no one in between can read or tamper with it.
WhatsApp	Confidentiality	End to End Encryption	Signal Protocol	Messages are encrypted on my devices before they even leave,and only the receiver's device can decrypt it.Even whatsapp's own server can't read chats.
Amazon	Authorization	Role-based Access Control(RBAC)	Https+Session Tokens	Once i log in to it,Amazon gives me a session token that decides what i'm allowed to do for ex:- like i can see my own order but not someone else's account or the admin dashboard
Netflix	Availability	Load Balancing+CDN	Content Delivery Network (CDN).Redundant servers	Netflix uses server spread across different locations,so even if one server goes down or there is a heavy traffic,i can still stream without the app crashing.

A brief explanation for each one of them:-

- **Gmail:-** It needs authentication because it's holding my personal emails,and anyone getting in would mean a huge privacy problem.
- **Google Pay:-** It needs both confidentiality and integrity since it's money-nobody should be able to see the transaction ,and nobody should be able to make changes to the amount in transit.
- **WhatsApp:** It is all about confidentiality as I don't want random people (or even)whatsapp itself)reading my personal chats.
- **Amazon:-** It needs proper authorization so that being logged in doesn't automatically mean ,I can access everything on the platform.

- **Netflix:-** It cares about availability because ,if the service keeps crashing,users would just leave and go to a competitor

2.Security Controls -Confidentiality,integrity,Availability

Here I picked the same five apps and looked at what controls they actually use and how these controls help before,during ,and after an attack.

Application	Control Used	Type(C/I/A)	Prevents	Detects	Recovers	Protocol/ Mechanism
Gmail	Multi-factor Authentication	Confidentiality	It stops any third party (attackers) from logging in even if they have my password	It sends a alert email if any login is attempted from a new/unknown device	It let me reset-password and log out of all other session	OTP/ Oauth2.0
Google Pay	TLS Encryption	Confidentiality&Integrity	It mainly stops hackers from reading or editing transaction data mid- transfer	User uses checksum and hash to detect if data was altered	Failed and tampered transaction are auto-rolled back	TLS,SSL, Certificate
WhatsApp	End to End Encryption	Confidentiality	It prevents anyone,including whatsapp from reading messages	N/A(E2E doesn't really "detect",it just blocks access	It is not really applicable since content is never exposed	Signal Protocol
Amazon	Firewalls+ intrusion Detection System(IDS)	Availability &Integrity	It Blocks suspicious traffic like DDos attempts before it even reaches the server	IDS flags unusual traffic pattern in real time	Backup servers take over so the site doesn't go down	Firewalls rules,IDS/ IPS systems

Netflix	Redundant servers +CDN	Availability	It is used to prevent single point failure one server crashing doesn't affect users	And the Monitoring tools detect server overload/downtime	Traffic is auto rerouted to a healthy server,so users barely notice	CDN,Load Balancers
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A short explanation of the CIA in my own word

Confidentiality is basically about data privacy,so apps like Gmail and WhatsApp use encryption so only the right people can read the infor/data.Integrity is about making sure data isn't tampered by someone in between, that's why Google Pay uses TLS with checksum,so if even a small bit data is tampered with during a transaction,it gets caught.Availability is about keeping the services up and running ,and that's mainly Netflix and Amazon's concern,since they need to handle huge traffic without crashing,that's where load balancing and redundant servers come in.

Also most of these controls work in a cycle

- **Prevention:**As the process pf stopping the attack before it even happens eg:encryption,firewalls,MFA.
- **Detection:**Catching it if it still happens eg:IDS,login alerts,monitoring tools.
- **Recovery:** the process of fixing things after the attack eg: backups,rollback,recruiting traffic.